



TEST YOURSELF 4

ANSWERS ON P. 574

1 Write each mixed numeral as an improper fraction.

a $1\frac{3}{4}$

b $2\frac{4}{9}$

c $5\frac{3}{5}$

d $3\frac{1}{7}$

2 Write each improper fraction as a mixed numeral.

a $\frac{7}{4}$

b $\frac{26}{3}$

c $\frac{23}{7}$

d $\frac{36}{24}$

3 Copy and complete each pair of equivalent fractions.

a $\frac{1}{3} = \frac{?}{15}$

b $\frac{3}{4} = \frac{12}{?}$

c $\frac{4}{5} = \frac{?}{20}$

d $\frac{5}{12} = \frac{15}{?}$

e $\frac{1}{2} = \frac{?}{50}$

f $\frac{3}{20} = \frac{?}{100}$

4 Simplify each fraction.

a $\frac{25}{30}$

b $\frac{30}{40}$

c $\frac{15}{20}$

d $\frac{20}{15}$

e $\frac{24}{48}$

f $\frac{24}{42}$

5 Order each set of fractions, from smallest to largest.

a $\frac{4}{5}, \frac{3}{10}, \frac{2}{5}, \frac{7}{10}$

b $\frac{1}{2}, \frac{1}{3}, \frac{7}{12}, \frac{11}{12}$

6 Evaluate each expression.

a $\frac{3}{4} + \frac{1}{4}$

b $\frac{2}{5} + \frac{3}{10}$

c $\frac{5}{10} - \frac{1}{3}$

d $\frac{3}{4} - \frac{1}{5}$

e $\frac{1}{2} + \frac{1}{3}$

f $\frac{1}{2} - \frac{3}{8}$

7 Evaluate each expression.

a $10 - \frac{1}{3}$

b $2\frac{5}{6} + 1\frac{1}{2}$

c $3\frac{2}{3} - 2\frac{5}{6}$

d $3 + 2\frac{1}{2} - 1\frac{1}{5}$

e $6 - 4\frac{2}{3}$

f $2\frac{5}{6} + 4\frac{1}{2} - 3\frac{2}{3}$

8 Evaluate each expression. (The letters are not in order.)

A $\frac{3}{7}$ of 21

B $\frac{2}{5}$ of 20

R $\frac{4}{9}$ of 27

O $\frac{1}{4}$ of 28

T $\frac{2}{3}$ of 30

H $\frac{1}{6}$ of 35

Order the 6 answers, beginning with the smallest. Replace each answer by its letter. What word do you spell?

4.01

4.01

4.01

4.01

4.02

4.03

4.04

4.05

4.05

9 Evaluate each expression.

- a** $\frac{2}{3}$ of 1 day (in hours) **b** $\frac{4}{5}$ of 2 km (in metres) **c** $\frac{3}{4}$ of 3 hours (in min)
d $\frac{7}{10}$ of 15 cm (in mm) **e** $\frac{1}{6}$ of 5 minutes (in seconds) **f** $\frac{5}{8}$ of 2 m (in cm)

4.06

10 Evaluate each product.

- a** $\frac{2}{3} \times \frac{1}{5}$ **b** $\frac{3}{5} \times 10$ **c** $\frac{4}{7} \times \frac{5}{8}$
d $\frac{1}{3} \times \frac{1}{7}$ **e** $1\frac{1}{2} \times \frac{1}{2}$ **f** $\frac{1}{5} \times 2\frac{1}{2}$
g $1\frac{1}{2} \times 1\frac{1}{2}$ **h** $2\frac{2}{7} \times 3\frac{1}{2}$ **i** $3\frac{1}{5} \times \frac{3}{8}$

4.07

11 Evaluate each quotient.

- a** $10 \div \frac{1}{5}$ **b** $\frac{1}{2} \div 3$ **c** $\frac{1}{2} \div \frac{1}{3}$
d $\frac{3}{4} \div \frac{3}{8}$ **e** $\frac{3}{5} \div \frac{1}{10}$ **f** $\frac{4}{7} \div \frac{5}{8}$
g $1\frac{1}{2} \div \frac{1}{2}$ **h** $2\frac{1}{4} \div 1\frac{1}{8}$ **i** $1\frac{7}{8} \div 1\frac{1}{4}$

4.08

12 The recipe for a fruit punch is 2 litres of ginger ale, $1\frac{1}{2}$ litres of orange juice, $2\frac{3}{4}$ litres of lemonade and $\frac{1}{3}$ of a litre of raspberry juice. How many litres of punch does this recipe make?

4.08

13 A farmer owns 24 goats. $\frac{1}{4}$ of the goats are black, $\frac{2}{3}$ are white, and the rest are divided equally between brown and spotted. How many of the goats are spotted?

4.09

14 Evaluate each expression.

- a** $\frac{13}{18} - \frac{3}{8}$ **b** $\frac{5}{6} + \frac{4}{5}$ **c** $3\frac{1}{4} - 2\frac{3}{4}$
d $\frac{9}{16} \times \frac{2}{45}$ **e** $\frac{17}{20} \div \frac{3}{5}$ **f** $2\frac{1}{3} \times 4\frac{2}{3}$

4.10

15 Convert each percentage to:**i** a fraction**a** 13%**b** 8%**ii** a decimal**c** $7\frac{1}{2}\%$

4.10

16 Convert each to a percentage.**a** $\frac{7}{10}$ **b** $\frac{23}{25}$ **c** $\frac{18}{80}$ **d** 0.8**e** 0.361**f** 0.04

- 17** Copy and complete this table.

Fraction	Decimal	Percentage
$\frac{3}{5}$		
	0.8	
		82%

4.10

- 18** Find:

- a** 40% of 90 **b** 12% of \$600 **c** 5% of 120
d 11% of \$210 **e** 34% of 150 kg **f** 65% of 360 m

4.11

- 19** In one hour of TV, 22% of the time was devoted to advertising. How many minutes were devoted to advertising?

4.11

- 20** The netball team scored 26 goals. Jessica scored 8 goals. What fraction of her team's score did Jessica make?

4.12

- 21** What percentage is 300 m of 1.5 km?

4.12

- 22** Anand scored 51 out of 60 for an Art project. What is his mark:

- a** as a simple fraction? **b** as a percentage?

4.12